

Elliott Bourgeois

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EDUCATION

Iowa State University

Ames, IA

Bachelor of Science in Electrical Engineering, Minor in German

Aug. 2023 – Anticipated May 2027

University Honors Program, President's List (6 semesters), Dean's List (6 semesters)

GPA: 4.0/4.0

Relevant Coursework: Analog VLSI Design; Microwave Circuit Design; Digital VLSI Design; Integrated Circuits; Antennas, Radiation, & Wave Propagation; Electromagnetism; Digital Signal Processing

EXPERIENCE

Analog IC Design Intern

May 2026 – Aug. 2026

Skyworks Solutions

Cedar Rapids, IA

- Supported transistor-level design and verification of analog blocks for Power Amplifier (PA) controller ICs across 180 nm and 55 nm CMOS and 40 nm SOI technologies.
- Modified and analyzed overvoltage/overcurrent protection, OTA, op-amp, and oscillator circuits, including porting existing designs across process technologies, using Cadence Virtuoso Schematic XL and ADE Assembler.
- Debugged circuits for stability, parasitics, headroom, and transistor operating-region issues through schematic and post-layout simulation using Cadence Virtuoso and Calibre PEX.
- Used PVT and Monte Carlo analysis to characterize performance and variation, guiding design decisions and preparing design reviews to communicate circuit performance and design tradeoffs.

Hardware Engineering Intern

May 2025 – Aug. 2025

Collins Aerospace

Cedar Rapids, IA

- Contributed to the design and validation of high-reliability Positioning, Navigation, and Timing (PNT) circuit cards, including high-frequency circuitry and GNSS/GPS subsystems.
- Developed specialized circuit card-based test fixtures and modular interfacing assemblies, accelerating integration and functional verification of RF, DSP, and antenna subsystems.

Undergraduate Research Assistant

Jan. 2024 – Jan. 2025

Iowa State University

Ames, IA

- Designed and validated custom PCBs for hardware system integration, supporting synchronized imaging with high-speed sensors and reducing hardware costs in a Photoacoustic Tomography platform.
- Engineered a programmable ESP32-based function generator in C++, producing arbitrary pulse trains configurable over UART.

Manufacturing Electrical Engineering Intern

May 2024 – Aug. 2024

Collins Aerospace

Coralville, IA

- Performed root cause analysis on complex PNT circuit card failures by interpreting waveforms and signal integrity issues using oscilloscopes, spectrum analyzers, multimeters, and automated test equipment (ATE).
- Built Python-based data analytics tools to optimize failure analysis and accelerate troubleshooting, reducing diagnostic cycle times by 30%.

EXTRACURRICULAR EXPERIENCE

Undergraduate Teaching Assistant – Digital Logic

Aug. 2025 – Present

Iowa State University - Department of Electrical and Computer Engineering

Ames, IA

- Leads labs and mentors students in Verilog, combinational and sequential logic, and implementation on FPGAs.
- Evaluates assignments and provides feedback to advance students' Verilog and digital design expertise.

TECHNICAL SKILLS

Languages: C, C++, Python, Verilog HDL, MATLAB, German (conversational)

Analog IC Design: Cadence Virtuoso Schematic XL, ADE Assembler, Layout XL, Calibre PEX, Genus, PVT & Monte Carlo Analysis

Design & Simulation Tools: Keysight ADS, Siemens Xpedition, KiCad, LTspice, NI Multisim, QuestaSim)

Test & Debug: Oscilloscopes, Waveform Generator, Spectrum Analyzers, Multimeters